Generalizing Behavior via IRL with Closed-Form Reward Centroids

Filippo Lazzati and Alberto Maria Metelli
{filippo.lazzati, albertomaria.metelli}@polimi.it

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Motivation and Contributions

Motivation

- **Transferring** behavior to new environments
- IRL is an **ill-posed** problem (Ng et al., 2000)

Contributions

- Novel **unbiased** criterion for reward selection
- **Closed-form** expression
- Provably efficient algorithms

Problem Setting

Feasible Sets

$$\mathcal{R}^{\mathrm{OPT},\overline{\mathcal{S}}}_{\mathcal{M},\pi_E} := \{r : \pi_E(s) \in \arg\max_a Q^*(s,a;r)\ \forall s \in \overline{\mathcal{S}}\}$$

$$\mathcal{R}^{\mathrm{MCE},\overline{\mathcal{S}}}_{\mathcal{M},\pi_E} := \{r : \pi_E(a|s) \propto e^{\frac{1}{\lambda}Q^*_\lambda(s,a;r)}\ \forall s \in \overline{\mathcal{S}}, \forall a\}$$

$$\mathcal{R}^{\mathrm{BIRL},\overline{\mathcal{S}}}_{\mathcal{M},\pi_E} := \{r : \pi_E(a|s) \propto e^{\frac{1}{\beta}Q^*(s,a;r)}\ \forall s \in \overline{\mathcal{S}}, \forall a\}$$

Problem Formulation

- Given $\mathcal{D}_E \sim \pi_E$
- Given $m \in \{\mathrm{OPT},\mathrm{MCE},\mathrm{BIRL}\}$
- Find:

$$\widehat{\pi}^* \in \arg\min_{\pi' \in \Pi_{c,k}} \Big(\max_{\pi \in \Pi_{c,k}} V^{\pi}(s'_0; r_E) - V^{\pi'}(s'_0; r_E) \Big),$$

where $\Pi_{c,k} = \{\pi \in \Pi : V^{\pi}(s'_0; c) \le k\}$.

Existing Methods

"Best-Case" Methods

(Ziebart et al., 2008)

Min suboptimality for any r from the feasible set:

$$\widehat{\pi}_{\mathrm{B}} \in \arg\min_{\pi' \in \Pi_{c,k}} \min_{r \in \mathcal{R}^m_{\mathcal{M},\pi_E}} \Big(\max_{\pi \in \Pi_{c,k}} V^{\pi}(s'_0; r) - V^{\pi'}(s'_0; r) \Big)$$

"Worst-Case" Methods

(Lazzati and Metelli, 2026)

Min suboptimality for worst $r \in \mathfrak{R} = [-1,+1]^{SA}$

$$\widehat{\pi}_{\mathrm{W}} \in \arg\min_{\pi' \in \Pi_{c,k}} \max_{r \in \mathcal{R}^m_{\mathcal{M},\pi_E} \cap \mathfrak{R}} \Big(\max_{\pi \in \Pi_{c,k}} V^{\pi}(s'_0; r) - V^{\pi'}(s'_0; r) \Big)$$

"On-Average" Methods

(Ramachandran and Amir, 2007)

Min average suboptimality w.r.t. $w : \mathbb{R}^{SA} \to [0,1]$

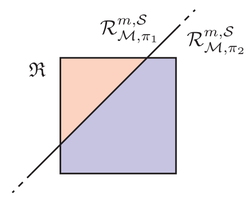
$$\widehat{\pi}_{w} \in \arg\min_{\pi' \in \Pi_{c,k}} \int_{\mathcal{R}^m_{\mathcal{M},\pi_E}} w(r) \Big(\max_{\pi \in \Pi_{c,k}} V^{\pi}(s'_0; r) - V^{\pi'}(s'_0; r) \Big) \mathrm{d}r$$

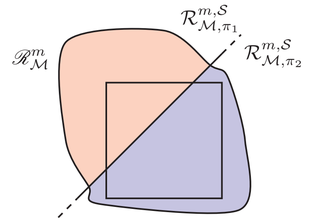
Limitations of Existing Methods

- "Best-Case" Methods → Overly **optimistic** in practice (Finn et al., 2016)
- "Worst-Case" Methods → Too **conservative** and computationally inefficient
- "On-Average" Methods → **Biased** in the space of policies for common $\overline{w}(r) := 1$ if $r \in \mathfrak{R} = [-1,+1]^{SA}$ else 0

Proposition 1 *There exists an MDP\R $\mathcal{M}$ and two deterministic policies $\pi_1, \pi_2 \in \Pi$ such that: $vol\big(\mathfrak{R} \cap \mathcal{R}^{\mathrm{OPT},\mathcal{S}}_{\mathcal{M},\pi_1}\big) \neq vol\big(\mathfrak{R} \cap \mathcal{R}^{\mathrm{OPT},\mathcal{S}}_{\mathcal{M},\pi_2}\big)$.*

Proposition 2 *There exists an MDP\R $\mathcal{M}$ and two stochastic policies $\pi_1, \pi_2 \in \Pi$ such that, for all $m \in \{MCE,BIRL\}$: $vol\big(\mathfrak{R} \cap \mathcal{R}^{m,\mathcal{S}}_{\mathcal{M},\pi_1}\big) \neq vol\big(\mathfrak{R} \cap \mathcal{R}^{m,\mathcal{S}}_{\mathcal{M},\pi_2}\big)$.*





Our Approach

Novel Priors over Rewards

$$\mathscr{R}^{\mathrm{OPT}}_{\mathcal{M}} := \{r \in \mathbb{R}^{SA} \mid \forall \pi \in \Pi^*(r), \forall s,a : |V^{\pi}(s;r)| \le C_1^{\mathrm{OPT}} k_\pi \wedge |A^{\pi}(s,a;r)| \le C_2^{\mathrm{OPT}}\},$$
$$\mathscr{R}^{\mathrm{MCE}}_{\mathcal{M}} := \{r \in \mathbb{R}^{SA} \mid \forall s,a : |V^*_\lambda(s;r)| \le C_1^{\mathrm{MCE}} \wedge |A^*_\lambda(s,a;r)| \le C_2^{\mathrm{MCE}}\},$$
$$\mathscr{R}^{\mathrm{BIRL}}_{\mathcal{M}} := \{r \in \mathbb{R}^{SA} \mid \forall s,a : |V^*(s;r)| \le C_1^{\mathrm{BIRL}} \wedge |A^*(s,a;r)| \le C_2^{\mathrm{BIRL}}\},$$

Proposition 3 ***(Close to hypercube)*** *Let $\mathcal{M}$ be any MDP\R. For some choices of the constants, then, for any $m \in \{OPT,MCE,BIRL\}$: $\mathfrak{R} \subseteq \mathscr{R}^m_{\mathcal{M}} \subseteq [-\rho C_1^m - C_2^m, +\rho C_1^m]^{SA}$.*

Proposition 4 ***(No bias)***

- *For any MDP\R $\mathcal{M}$ and any pair of deterministic policies $\pi_1, \pi_2 \in \Pi$, it holds that: $vol(\mathscr{R}^{OPT}_{\mathcal{M}} \cap \mathcal{R}^{OPT,\mathcal{S}}_{\mathcal{M},\pi_1}) = vol(\mathscr{R}^{OPT}_{\mathcal{M}} \cap \mathcal{R}^{OPT,\mathcal{S}}_{\mathcal{M},\pi_2})$.*
- *For any MDP\R $\mathcal{M}$ and any pair of stochastic policies $\pi_1, \pi_2 \in \Pi$ such that $\pi_1(a|s) \ge \pi_{\min}$ and $\pi_2(a|s) \ge \pi_{\min}$ in every s,a, if $C_2^{MCE} \ge \lambda \log(1/\pi_{\min})$, then: $vol(\mathscr{R}^{MCE}_{\mathcal{M}} \cap \mathcal{R}^{MCE,\mathcal{S}}_{\mathcal{M},\pi_1}) = vol(\mathscr{R}^{MCE}_{\mathcal{M}} \cap \mathcal{R}^{MCE,\mathcal{S}}_{\mathcal{M},\pi_2})$.*
- *For any MDP\R $\mathcal{M}$ and any pair of stochastic policies $\pi_1, \pi_2 \in \Pi$ such that $\pi_1(a|s) \ge \pi_{\min}$ and $\pi_2(a|s) \ge \pi_{\min}$ in every s,a, if $C_2^{BIRL} \ge \beta \log(1/(A\pi_{\min}))$, then: $vol(\mathscr{R}^{BIRL}_{\mathcal{M}} \cap \mathcal{R}^{BIRL,\mathcal{S}}_{\mathcal{M},\pi_1}) = vol(\mathscr{R}^{BIRL}_{\mathcal{M}} \cap \mathcal{R}^{BIRL,\mathcal{S}}_{\mathcal{M},\pi_2})$.*

Closed-form Expressions and Algorithms

Algorithm 1: OPT

Input: Data $\mathcal{D}_E$
$\widehat{\mathcal{S}} \leftarrow$ visited states
$\widehat{\mathcal{Z}} \leftarrow$ visited state-action pairs
$$\widehat{r}^{\mathrm{OPT}}(s,a) \leftarrow \begin{cases} 1 & \text{if } (s,a) \in \widehat{\mathcal{Z}} \\ \frac{1}{A} & \text{if } s \notin \widehat{\mathcal{S}} \\ 0 & \text{otherwise} \end{cases}$$
Return $\widehat{r}^{\mathrm{OPT}}$

Algorithm 2: MCE

Input: Data $\mathcal{D}_E$, parameter $\pi'_{\min}$
$N(s,a) \leftarrow$ first-time state-action counts
$N(s) \leftarrow \sum_{a \in \mathcal{A}} N(s,a)$
$$\widehat{\pi}(a|s) \leftarrow \max\Big\{\pi'_{\min}, \frac{N(s,a)}{\max\{1, N(s)\}}\Big\}$$
$\widehat{r}^{\mathrm{MCE}}(s,a) \leftarrow \log \widehat{\pi}(a|s)$
Return $\widehat{r}^{\mathrm{MCE}}$

Algorithm 3: BIRL

Input: Data $\mathcal{D}_E$, parameter $\pi'_{\min}$
$N(s,a) \leftarrow$ first-time state-action counts
$N(s) \leftarrow \sum_{a \in \mathcal{A}} N(s,a)$
$$\widehat{\pi}(a|s) \leftarrow \max\Big\{\pi'_{\min}, \frac{N(s,a)}{\max\{1, N(s)\}}\Big\}$$
$$\widehat{r}^{\mathrm{BIRL}}(s,a) \leftarrow \begin{cases} \log \frac{\widehat{\pi}(a|s)}{\max_{a'} \widehat{\pi}(a'|s)} & \text{if } N(s) > 0, \\ \log \pi'_{\min} & \text{if } N(s) = 0. \end{cases}$$
Return $\widehat{r}^{\mathrm{BIRL}}$

Sample Complexity Analysis

Theorem 5 *Let $\delta \in (0,1)$ and $H \ge S$. If π_E is deterministic, then, with probability at least $1-\delta$, it holds that $\widehat{r}^{OPT} = r^{OPT}_{\mathcal{M},\pi_E}$ provided that:*

$$NH \ge \frac{H \log \frac{|\mathcal{S}_{\mathcal{M},\pi_E}|}{\delta}}{p^{\min,H}_{\mathcal{M},\pi_E}}.$$

where $p^{\min,H}_{\mathcal{M},\pi_E}$ is the minimum probability with which π_E visits any state in its support in the first H steps.

Theorem 6 *Let $\epsilon, \delta \in (0,1)$ and $H \ge S$. If $C_2^{MCE} \ge \lambda \log(1/\pi_{\min})$ and $\pi_{\min} \ge \pi'_{\min}$, then, with probability $1-\delta$, we have $\|\widehat{r}^{MCE} - r^{MCE}_{\mathcal{M},\pi_E}\|_\infty \le \epsilon$ provided that:*

$$NH \ge \frac{H \log^2 \frac{4SA}{\delta}}{p^{\min,H}_{\mathcal{M},\pi_E}} \cdot \frac{16}{\epsilon^2 \pi'^2_{\min}}.$$

Theorem 7 *Let $\epsilon, \delta \in (0,1)$ and $H \ge S$. If if $C_2^{BIRL} \ge \beta \log(1/(A\pi_{\min}))$ and $\pi_{\min} \ge \pi'_{\min}$, then, with probability $1-\delta$, we have $\|\widehat{r}^{BIRL} - r^{BIRL}_{\mathcal{M},\pi_E}\|_\infty \le \epsilon$ provided that:*

$$NH \ge \frac{H \log^2 \frac{8SA}{\delta}}{p^{\min,H}_{\mathcal{M},\pi_E}} \cdot \frac{33}{\epsilon^2 \pi'^2_{\min}}.$$

Open Questions

- How to scale in a provably efficient manner to large/continuous state-action spaces?
- Empirical study in real-world environments, e.g., financial domains
- Simulations to compare practical (e.g., deep) implementations of RS-BC with vanilla BC

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